

10.6 Exploring parabolas



One of the simplest and most important non-linear graphs is the parabola. When a ball is thrown or water streams up and out from a garden hose or fountain, the path followed has a parabolic shape. The parabola is the graph of a quadratic relation with the basic rule $y = x^2$. Quadratic rules such as $y = (x - 1)^2$ and $y = 2x^2 - x - 3$ also give graphs that are parabolas and are transformations of the graph of $y = x^2$.

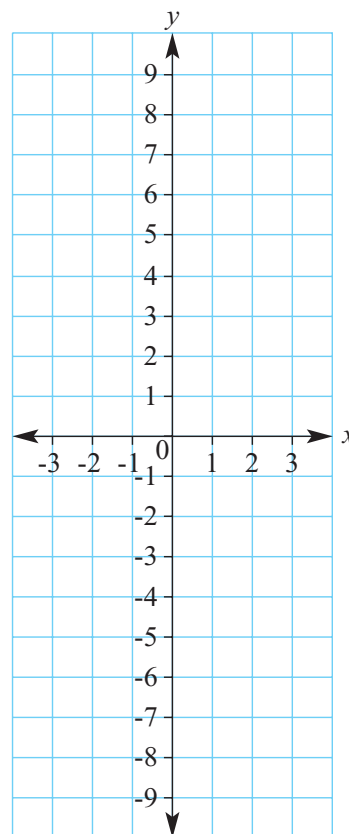


► Let's start: To what effect?

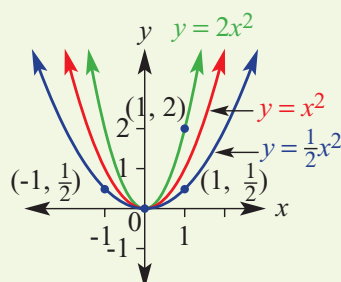
To see how different quadratic rules compare to the graph of $y = x^2$, complete this table and plot the graph of each equation on the same set of axes.

x	-3	-2	-1	0	1	2	3
$y_1 = x^2$	9	4					
$y_2 = -x^2$	-9						
$y_3 = (x - 2)^2$	25	16	9				
$y_4 = x^2 - 3$	6						

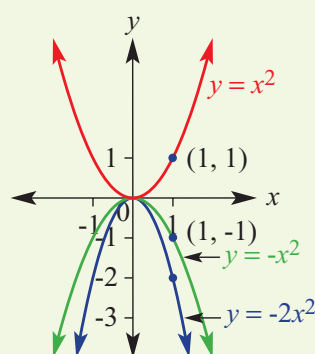
- For all the graphs, find such features as the:
 - turning point
 - axis of symmetry
 - y -intercept
 - x -intercepts.
- Discuss how each of the graphs of y_2 , y_3 and y_4 compare to the graph of $y = x^2$. Compare the rule with the position of the graph.



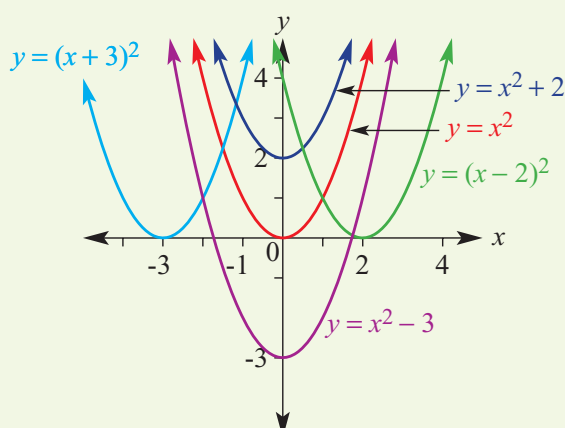
- A **parabola** is the graph of a quadratic relation. The basic parabola has the rule $y = x^2$.
 - The vertex (or turning point) is $(0, 0)$.
 - It is a minimum turning point.
 - Axis of symmetry is $x = 0$.
 - y -intercept is $(0, 0)$.
 - x -intercept is $(0, 0)$.
- Simple transformations of the graph of $y = x^2$ include:
 - **dilation**



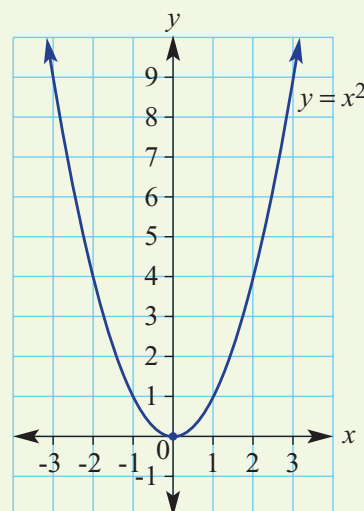
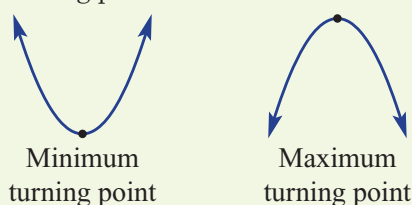
- **reflection**



- **translation**



- Turning points can be a maximum or a minimum.



Parabola

A smooth U-shaped curve with the basic rule $y = x^2$

Dilation

A transformation where a curve is enlarged or reduced but the centre is not changed

Reflection

A transformation where a curve is flipped across a line on the number plane

Translation

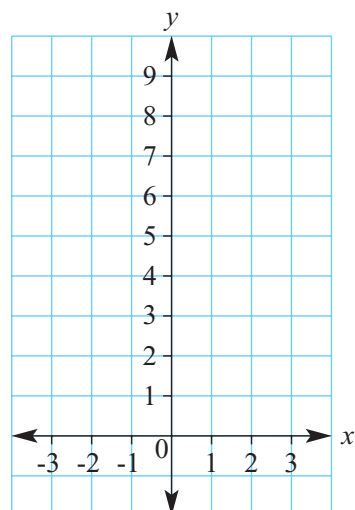
A transformation where a curve is moved a certain distance on the number plane

Exercise 10F

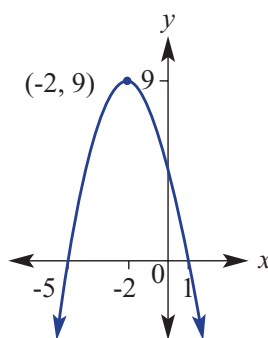
Understanding

- 1 Complete this table and grid to plot the graph of $y = x^2$.

x	-3	-2	-1	0	1	2	3
y	9						



- 2 Write the missing features for this graph.



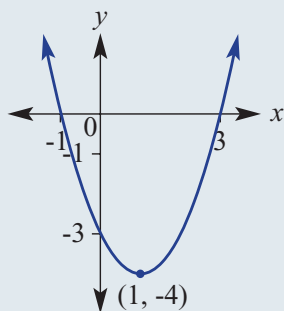
- The parabola has a _____ (maximum or minimum).
- The coordinates of the turning point are _____.
- The coordinates of the y -intercept are _____.
- The coordinates of the x -intercepts are _____ and _____.
- The axis of symmetry is _____.

Example 16 Identifying key features of parabolas

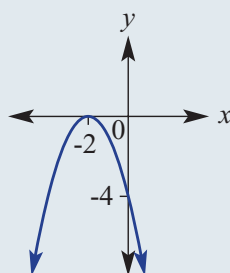
Determine the following key features of each of the given graphs.

- Turning point and whether it is a maximum or minimum
- Axis of symmetry
- x -intercepts
- y -intercept

a



b



Solution

Explanation

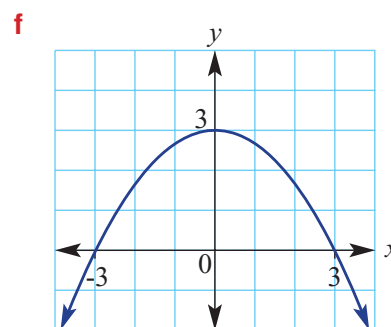
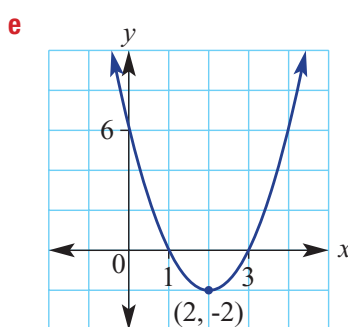
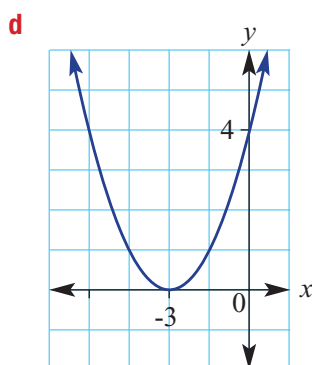
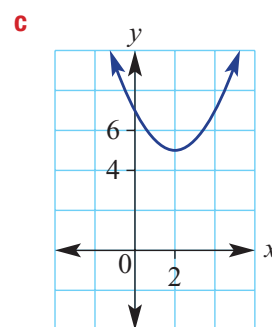
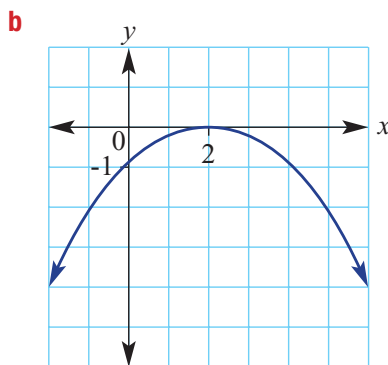
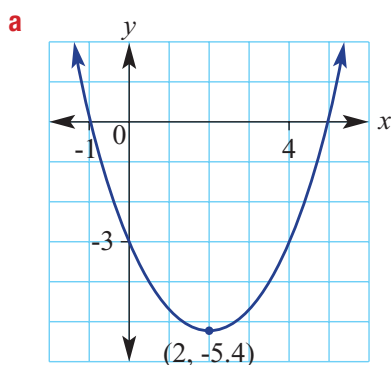
- a**
- i** Turning point is a minimum at $(1, -4)$.
 - ii** Axis of symmetry is $x = 1$.
 - iii** x -intercepts at $(-1, 0)$ and $(3, 0)$.
 - iv** y -intercept at $(0, -3)$.
- b**
- i** Turning point is a maximum at $(-2, 0)$.
 - ii** Axis of symmetry is $x = -2$.
 - iii** x -intercept at $(-2, 0)$.
 - iv** y -intercept at $(0, -4)$.

Lowest point of graph is at $(1, -4)$.
 Line of symmetry is through the x -coordinate of the turning point.
 x -intercepts lie on the x -axis ($y = 0$) and the y -intercept on the y -axis ($x = 0$).
 Graph has a highest point at $(-2, 0)$.
 Line of symmetry is through the x -coordinate of the turning point.
 Turning point is also the one x -intercept.

3 Determine these key features of the following graphs.

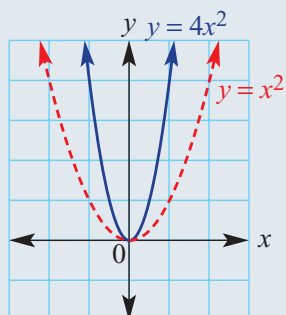
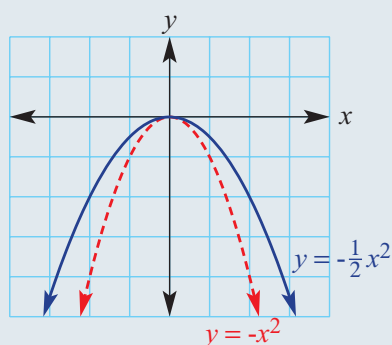
- i** turning point and whether it is a maximum or minimum
- ii** axis of symmetry
- iii** x -intercepts
- iv** y -intercept

An axis of symmetry is described by a rule such as $x = 2$ or $x = -3$.



Example 17 Dilating and reflecting parabolas

Copy and complete the table for the following graphs.

a**b****a****b**

	Formula	Maximum or minimum	Reflected in the x-axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = 4x^2$					
b	$y = -\frac{1}{2}x^2$					

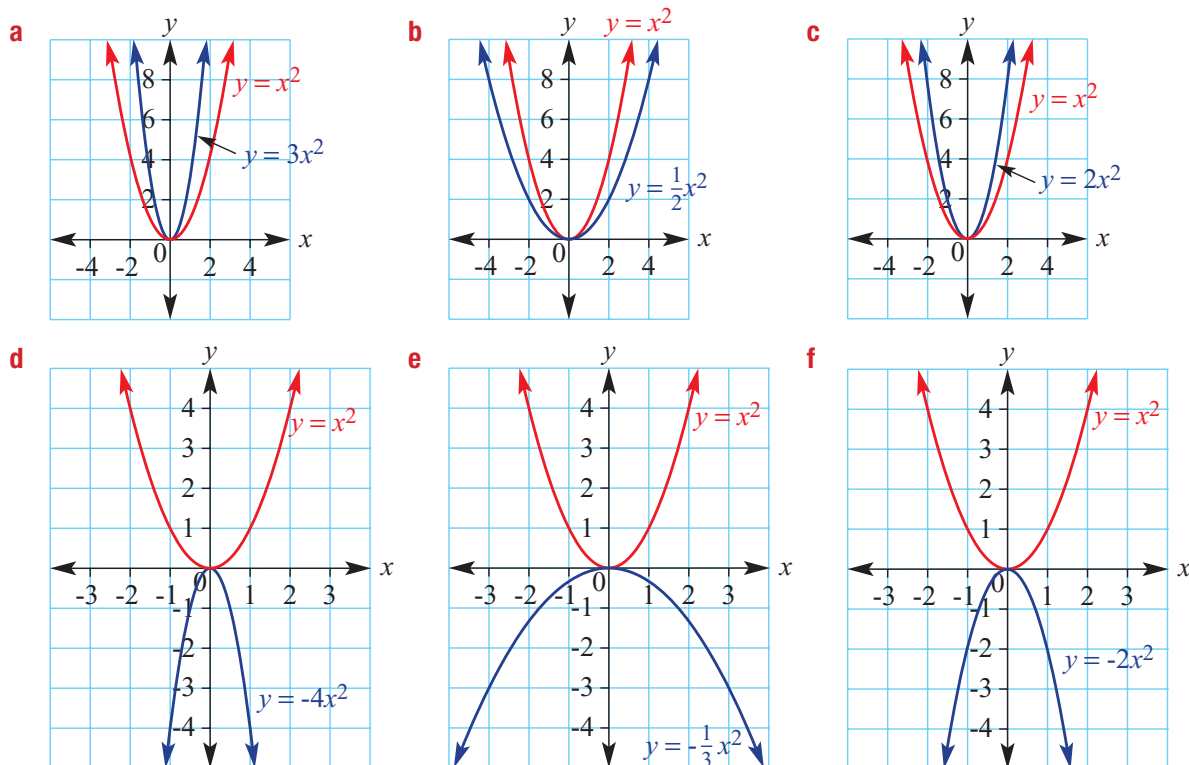
Solution**Explanation**

	Formula	Maximum or minimum	Reflected in the x-axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = 4x^2$	Minimum	No	(0, 0)	4	Narrower
b	$y = -\frac{1}{2}x^2$	Maximum	Yes	(0, 0)	$-\frac{1}{2}$	Wider

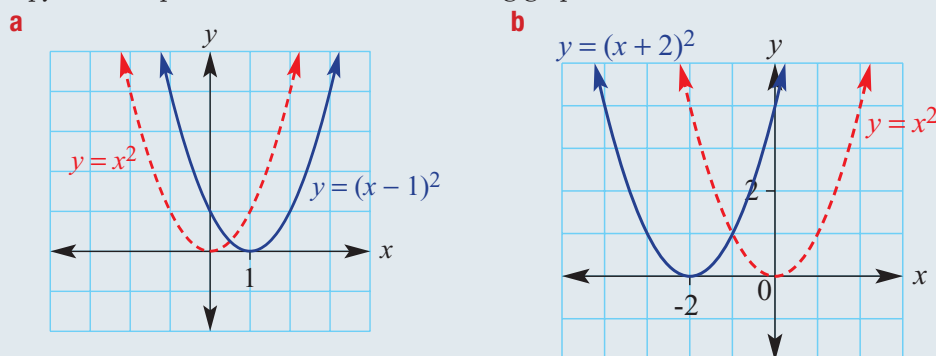
Read features from graphs and consider the effect of each change in equation on the graph.

4 Copy and complete the table below for the graphs that follow.

	Formula	Maximum or minimum	Reflected in the x-axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = 3x^2$					
b	$y = \frac{1}{2}x^2$					
c	$y = 2x^2$					
d	$y = -4x^2$					
e	$y = -\frac{1}{3}x^2$					
f	$y = -2x^2$					


Example 18 Translating horizontally

Copy and complete the table for the following graphs.



	Formula	Maximum or minimum	Reflected in the x -axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = (x - 1)^2$					
b	$y = (x + 2)^2$					

Solution
Explanation

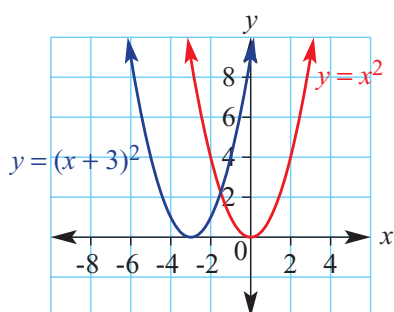
	Formula	Maximum or minimum	Reflected in the x -axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = (x - 1)^2$	Minimum	No	(1, 0)	0	Same
b	$y = (x + 2)^2$	Minimum	No	(-2, 0)	9	Same

 The effect is to shift right or left: Right for part **a** and left for part **b**.

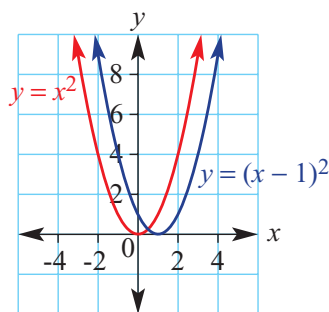
5 Copy and complete the table below for the graphs that follow.

	Formula	Turning point	Axis of symmetry	y-intercept ($x = 0$)	x-intercept
a	$y = (x + 3)^2$				
b	$y = (x - 1)^2$				
c	$y = (x - 2)^2$				
d	$y = (x + 4)^2$				

a



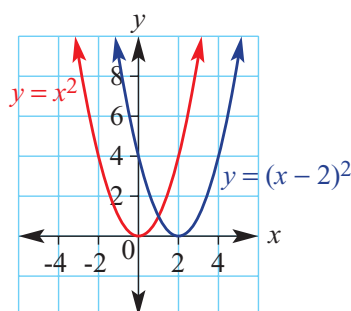
b



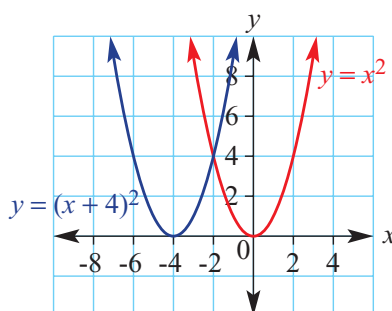
The axis of symmetry is a vertical line passing through the turning point of a parabola. The equation is given by the x coordinate of the turning point.



c



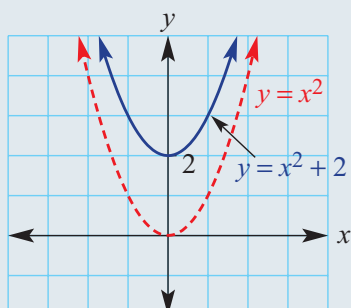
d



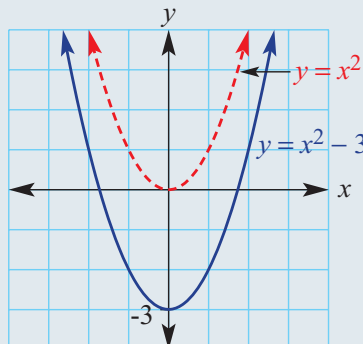
Example 19 Translating vertically

Copy and complete the table for the following graphs.

a



b



	Formula	Maximum or minimum	Reflected in the x -axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = x^2 + 2$					
b	$y = x^2 - 3$					

Solution

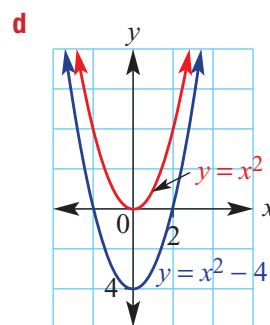
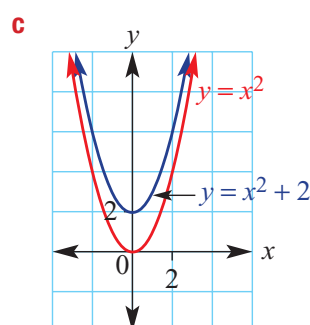
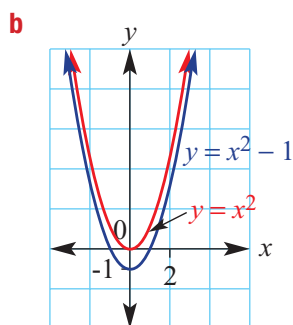
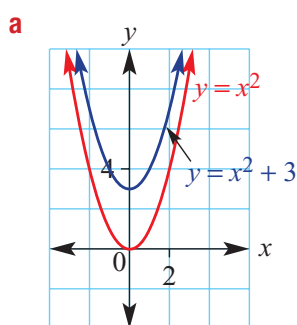
Explanation

	Formula	Maximum or minimum	Reflected in the x -axis (yes/no)	Turning point	y value when $x = 1$	Wider or narrower than $y = x^2$
a	$y = x^2 + 2$	Minimum	No	(0, 2)	3	Same
b	$y = x^2 - 3$	Minimum	No	(0, -3)	-2	Same

The effect is to shift up or down: Up for $y = x^2 + 2$ and down for $y = x^2 - 3$.

6 Copy and complete the table for the graphs that follow.

	Formula	Turning point	y -intercept ($x = 0$)	y value when $x = 1$
a	$y = x^2 + 3$			
b	$y = x^2 - 1$			
c	$y = x^2 + 2$			
d	$y = x^2 - 4$			



Problem-solving and Reasoning

7 Match each of the following equations to one of the graphs below.

a $y = 2x^2$

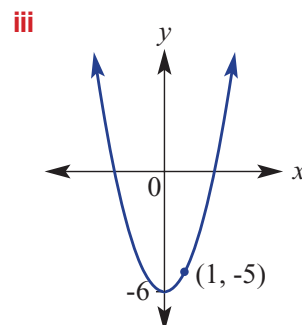
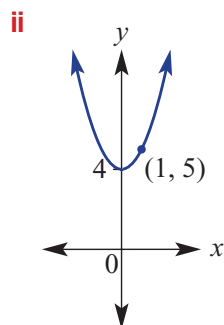
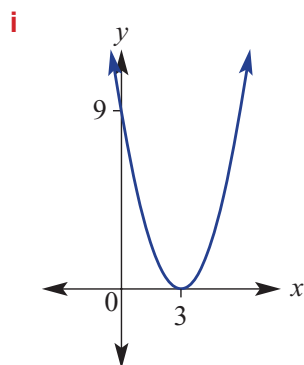
b $y = x^2 - 6$

c $y = (x + 2)^2$

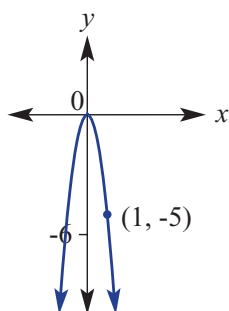
d $y = -5x^2$

e $y = (x - 3)^2$

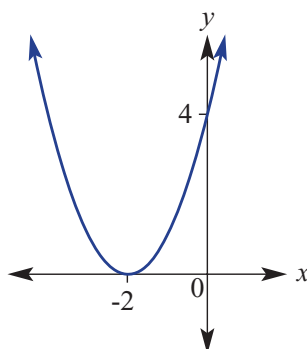
f $y = x^2 + 4$



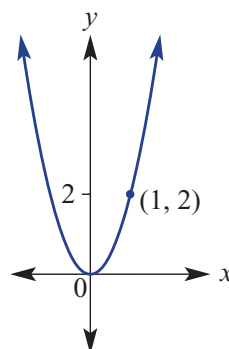
iv



v



vi



8 Write a rule for a parabola with each feature.

- a Same shape as $y = x^2$, minimum turning point $(0, 2)$
- b Same shape as $y = x^2$, maximum turning point $(0, 0)$
- c Same shape as $y = x^2$, minimum turning point $(-1, 0)$
- d Same shape as $y = x^2$, minimum turning point $(5, 0)$

What turns $y = x^2$ into a graph with a maximum turning point?



Parabolas with technology



9 a Using technology, plot the following pairs of graphs on the same set of axes for $-5 \leq x \leq 5$ and compare their tables of values.

- i $y = x^2$ and $y = 4x^2$
- ii $y = x^2$ and $y = \frac{1}{3}x^2$
- iii $y = x^2$ and $y = 6x^2$
- iv $y = x^2$ and $y = \frac{1}{4}x^2$
- v $y = x^2$ and $y = 7x^2$
- vi $y = x^2$ and $y = \frac{2}{5}x^2$

b Suggest how the constant a in $y = ax^2$ transforms the graph of $y = x^2$.



10 a Using technology, plot the following sets of graphs on the same set of axes for $-5 \leq x \leq 5$ and compare the turning point of each.

- i $y = x^2, y = (x + 1)^2, y = (x + 2)^2, y = (x + 3)^2$
- ii $y = x^2, y = (x - 1)^2, y = (x - 2)^2, y = (x - 3)^2$

b Explain how the constant b in $y = (x + b)^2$ transforms the graph of $y = x^2$.



11 a Using technology, plot the following sets of graphs on the same set of axes for $-5 \leq x \leq 5$ and compare the turning point of each.

- i $y = x^2, y = x^2 + 1, y = x^2 + 2, y = x^2 + 3$
- ii $y = x^2, y = x^2 - 1, y = x^2 - 3, y = x^2 - 5$

b Explain how the constant k in $y = x^2 + k$ transforms the graph of $y = x^2$.



Using technology 10.6: Sketching parabolas

This activity is available on the companion website as a printable PDF.